Quiz 2 — Sample 2

DATA 412/612

Write your answers in an R script. Use the tidyverse. Load `nycflights13` and the `flights` data. This is a second practice quiz. It is not a solution key.

1. In `flights`, for every `carrier`, find the number of `NA`s in each column that is numeric and starts with `dep`, or that is character and contains `er` or `il`.

2. Restrict attention to winter flights (December, January, or February) with `air_time` greater than 180 minutes. Find the carrier that operated the fewest of these flights. Then, for every `origin`, count how many of that carrier's winter long flights departed from that origin.

3. Among origin–destination pairs with at least 100 flights, which pair has the highest average `distance`? Report the origin, the destination, the number of flights, and the average distance.

4. Which three months have the highest proportion of flights with `arr_delay` greater than 30 minutes? Report the month, the number of flights, the number of those flights that were delayed, and the proportion. When you count delayed flights, ignore missing `arr_delay`. Keep every flight in the denominator.

5. Use `readr` parse functions to correct the type of each column below. The column names are a hint.

   - `date` is year, month, day, with `..` between the pieces. The string `NA` means missing.
   - `time` is hours, minutes, seconds, with `..` between the pieces.
   - `num` is a number that may contain commas. The string `n/a` means missing.
   - `log` is logical. The string `missing` means missing.
   - `char` is stored as a factor. It should end up as character. The string `miss` means missing.
   - `fac` should be a factor.

```
df <- data.frame(
  date = c("2024..03..15", "2022..12..01", "NA"),
  time = c("09..15..00", "21..40..30", "07..05..05"),
  num  = c("1,200", "850", "n/a"),
  log  = c("TRUE", "missing", "FALSE"),
  char = as.factor(c("red", "blue", "miss")),
  fac  = c("X", "Y", "Z")
)
```